

WATERFALL Vs. AGILE DEVELOPMENT



Unique AI Projects Challenges



- AI Projects offer unique challenges compared to traditional software projects.

Uncertain and evolving requirements:

AI project goals often shift based on data insights and model performance, making upfront planning difficult.

Data as a core dependency: Success heavily depends on data availability, quality, and relevance.

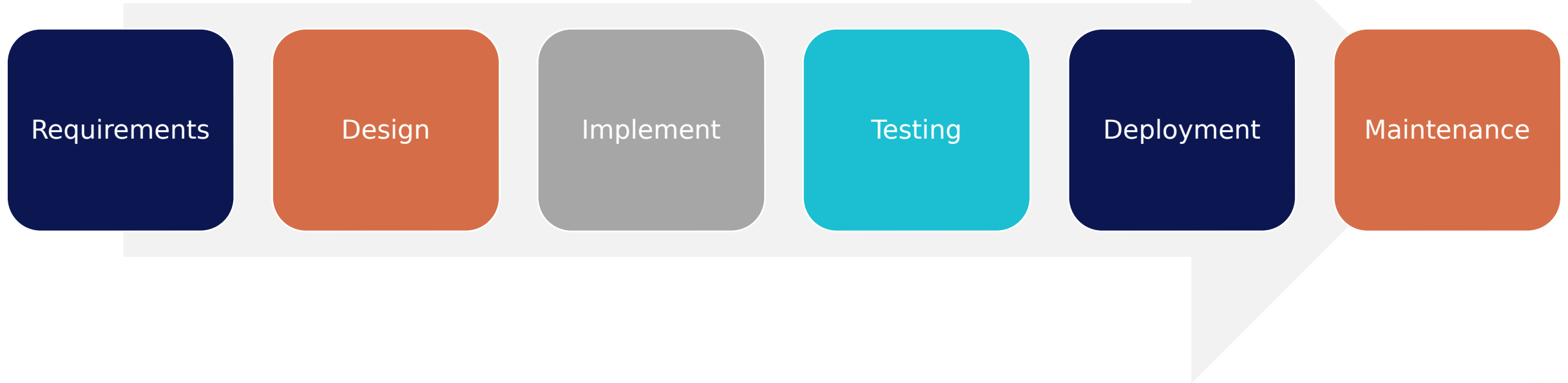
Non-deterministic outcomes: Unlike traditional software, AI models may never reach perfect accuracy, requiring tolerance for probabilistic results.

Complex team collaboration: AI projects require cross-functional teams (data scientists, engineers, domain experts), making coordination and communication more challenging.



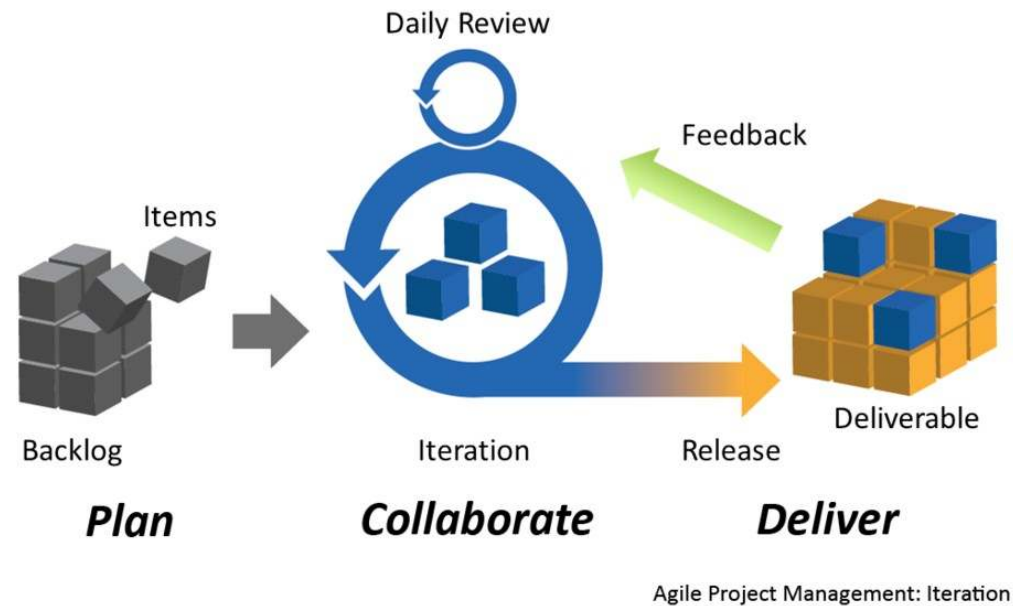
Waterfall Development

- Traditional Waterfall development is a linear and sequential software development methodology where each phase must be completed before the next one begins.
- This rigid approach allows minimal flexibility for changes once development starts.
- Stakeholders don't see the final product until all phases are complete, making it best for well-defined, predictable projects.
- Example: Building a bridge. The entire design is finalized upfront, construction happens once, and testing is conducted only at the end.



Agile Development

- Agile development is a flexible, iterative approach to building products that focuses on collaboration, continuous feedback, and delivering small, working pieces frequently.
- AI Projects are exploratory; the exact outcome can be uncertain, evolving with data/experimentation.
- Work is broken into smaller, manageable increments delivered in short cycles, allowing for rapid feedback and adaptation.



Waterfall Vs. Agile Development



Feature / Aspect	Agile Development	Waterfall Development
Approach	Iterative and incremental	Linear and sequential
Flexibility	Highly flexible; changes can be made at any stage	Inflexible; changes are difficult once a phase is completed
Project Phases	Phases (planning, design, development) are repeated in each iteration	Phases are distinct and completed one after the other
Customer Involvement	Continuous involvement and feedback	Limited involvement, mostly at the beginning and end
Testing	Ongoing throughout the development cycle	Done after the development phase
Delivery	Working product delivered in increments	Delivered at the end of the project
Risk Management	Lower risk due to frequent reassessment and delivery	Higher risk due to late testing and feedback
Team Collaboration	Strong emphasis on collaboration and self-organizing teams	Typically follows a top-down approach
Project Size & Complexity	Suitable for complex, evolving projects	Suitable for simple, well-defined projects
Examples of Use	Software startups, evolving products, dynamic requirements	Construction projects, manufacturing, regulated environments



AI PROJECT MANAGEMENT FRAMEWORKS: CRISP-DM Vs. AGILE Vs. HYBRID





A. CRISP-DM

A classic data science methodology

Business understanding →
Data understanding → Data
preparation → Modeling →
Evaluation → Deployment.

B. Agile for AI

Modified Agile practices (e.g., sprints, stand-ups) tailored for iterative experimentation, rapid prototyping, and model tuning. Very common in AI/ML teams.

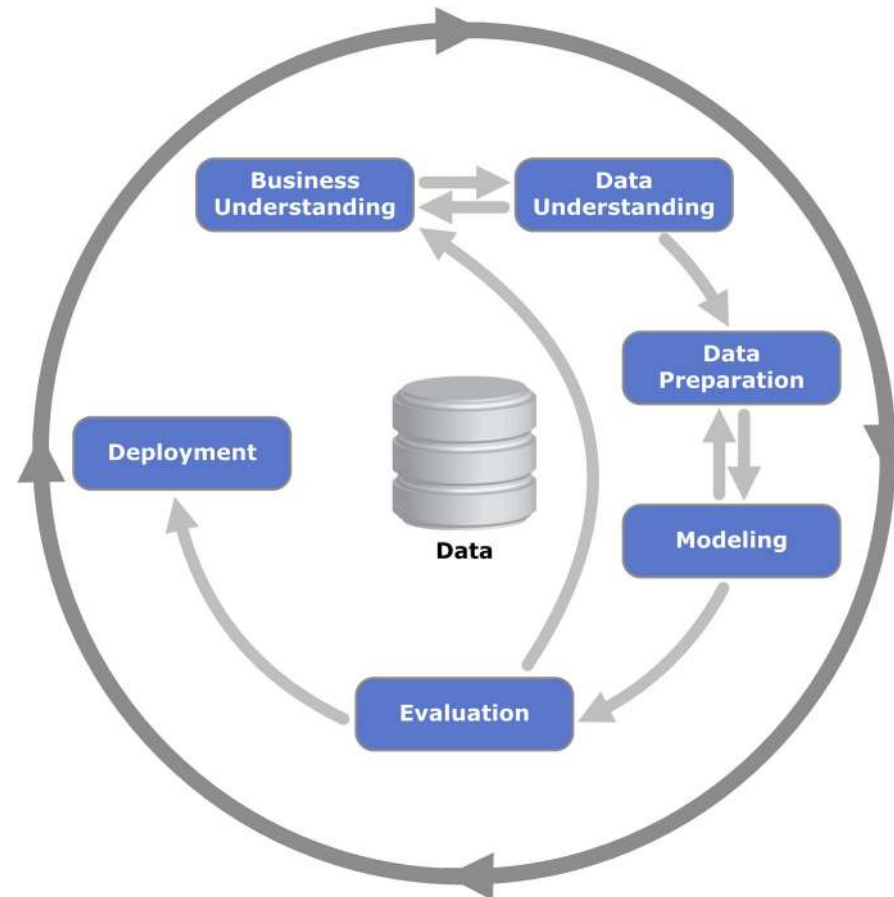
C. Hybrid Approaches

Combining traditional project management with Agile and CRISP-DM to adapt to evolving data and model needs. Helps bridge business goals and technical execution.



A. Crisp-DM Approach

- CRISP-DM stands for Cross-Industry Standard Process for Data Mining. It's one of the most widely used and foundational methodologies for managing data science and AI projects.
- It provides a structured, six-phase process to guide teams from business understanding to deploying a working solution.



A. Crisp-DM Phases



Phase	What Happens
1. Business Understanding	Define the project goals, success criteria, and business objectives. Understand the problem deeply.
2. Data Understanding	Collect initial data, explore it, and identify quality issues, patterns, or interesting features.
3. Data Preparation	Clean, transform, and engineer features. This step often takes the most time (up to 80%).
4. Modeling	Choose algorithms, build models, and experiment with parameters.
5. Evaluation	Assess how well the model meets business goals, not just accuracy. Ensure it solves a real problem.
6. Deployment	Deliver the model into production. May include documentation and monitoring.



B. Agile for AI

- Agile for AI is an adaptation of Agile project management tailored to the needs of AI/ML Projects.
- It focuses on building AI solutions iteratively, collaborating across teams, and responding quickly to changes in data, performance, or business goals.
- AI projects are uncertain by nature; you don't always know what data will work, which model will perform best, or how users will react. Agile allows teams to explore, learn, and pivot quickly rather than following a rigid, linear plan.

Short, iterative sprints to develop, test, and refine models.

Continuous collaboration between data scientists, engineers, and business teams.

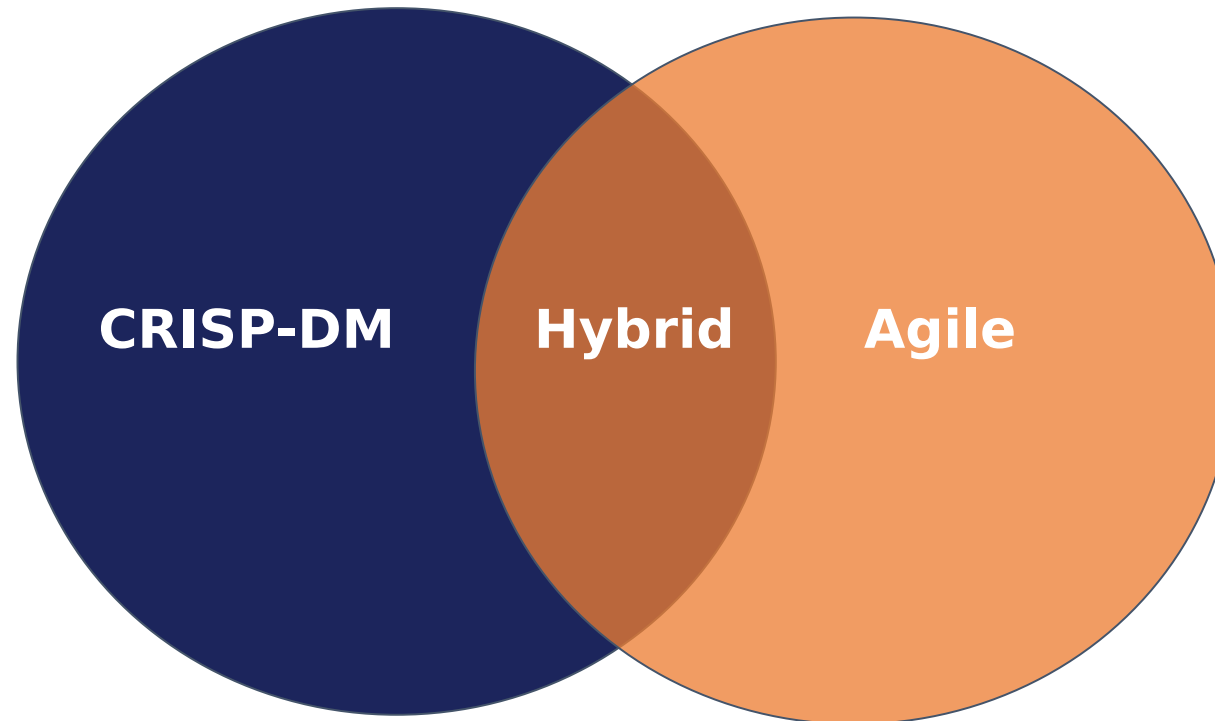
Fast feedback loops to validate models early and improve performance over time.

Flexibility to adjust the approach as new data or challenges emerge.



C. Hybrid Approaches

- A Hybrid Approach combines the structured phases of CRISP-DM with the flexibility and speed of Agile to manage AI projects more effectively.
- It brings the best of both worlds: clear planning and documentation plus rapid, iterative development.
 - Structured Phases with Flexibility: Follows CRISP-DM's stages (e.g., Business Understanding, Data Preparation, Modeling) but applies Agile sprints within each phase for quick iteration.
 - Milestones + Sprints: Project progress is tracked using clear milestones (good for reporting) while Agile sprints enable fast prototyping and experimentation in parallel.
 - Balanced Team Workflow: Supports both regulated environments (that need documentation and control) and dynamic teams (that need to test and adapt quickly).



Comparison Summary



Aspect	CRISP-DM	Agile for AI	Hybrid Approach
Origin	Classic data science methodology	Software development (adapted for AI)	Blends CRISP-DM with Agile and traditional PM frameworks
Structure	Linear and phase-based	Iterative and cyclical	Structured milestones + iterative sprints within each phase
Focus	Comprehensive project phases	Rapid experimentation and delivery	Aligning long-term planning with short-term agility
Planning Style	Heavy up-front planning	Light planning, frequent adjustments	Initial high-level roadmap + flexible sprint planning
Team Interaction	Stakeholder involvement mostly at key milestones	Frequent collaboration (daily stand-ups, sprint reviews)	Regular sprint reviews + milestone-based stakeholder check-ins
Speed of Feedback	Slower (after full phase completion)	Fast (every sprint or 1-2 weeks)	Medium speed: sprint-based feedback but tied to business milestone reviews
Example Use	Structured projects with fixed scope	Projects needing rapid prototyping and tuning	Enterprise-level AI projects, regulated industries, or cross-functional efforts
Best For	Research, stable environments	Startups, evolving problem spaces	Large, strategic AI initiatives that need both control and flexibility
Risks Addressed	Misalignment with real-time changes	Lack of structure or documentation	Combines governance with adaptability



Example

- JP Morgan Chase wants to use AI to predict which customers are most likely to default on their loans. The objective is to improve risk management, automate loan approval decisions, and comply with regulatory standards.
- Let's explore the 3 approaches: CRISP-DM, Agile and Hybrid.



<https://www.flickr.com/photos/bensutherland/178395814/in/photostream/>

CRISP-DM Approach



1. Business Understanding: Stakeholders want to reduce default-related losses and automate decisions for faster loan approvals.
2. Data Understanding: Explore data from past loan applicants, including demographics, employment history, credit scores, loan amounts, and repayment history.
3. Data Preparation: Clean missing values, encode categorical features (e.g., employment type), and create engineered features like debt-to-income ratio or credit utilization rate.
4. Modeling: Train classification models like XGBoost. Use cross-validation and grid search for hyperparameter tuning.
5. Evaluation: Use metrics like precision, recall, and F1-score. Evaluate fairness and bias (e.g., by gender or income level) using SHAP values and bias audits.
6. Deployment: Deploy the model through the bank's loan approval portal. Monitor model drift and performance with batch scoring and dashboards.





1. Sprints:

Week	Sprint Goals	Activities	Deliverables
Week 1	Sprint 0: Planning & Setup	- Define problem and success metrics - Align with stakeholders - Set up data access & infrastructure	Product backlog, user stories, and sprint plan
Week 2	Sprint 1: Data Exploration & Baseline	- Explore data - Identify data quality issues - Build and evaluate baseline model	Data profiling report, baseline model results
Week 3	Sprint 2: Feature Engineering & Model Iteration	- Engineer features - Tune hyperparameters - Try multiple models	Improved model performance, Jupyter notebooks
Week 4	Sprint 3: Evaluation & Explainability	- Test for fairness & bias - Add explainability (SHAP, LIME) - Validate with business team	Evaluation metrics, explainability report
Week 5	Sprint 4: Deployment & Integration	- Prepare model for production - Set up model serving pipeline - Integrate with application	Model API, deployment script
Week 6	Sprint 5: Monitoring & Feedback Loop	- Set up dashboards - Collect post-deployment feedback - Plan for retraining triggers	Monitoring dashboard, final report

2. Daily Stand-ups (15-mins):

Data scientists, product owners, and compliance officers discuss blockers like missing data or fairness concerns.

3. Rapid Experimentation:

Test multiple models, sampling strategies, and explainability tools in short cycles.



Hybrid Approach



- **Traditional PM Backbone:**
Define clear milestones (e.g., data readiness, model MVP, compliance approval, deployment) and Gantt timelines to satisfy senior management and regulators.
- **Agile Loops within Milestones:**
Use Agile cycles within each phase. For example, iterate on feature engineering while following CRISP-DM's structured flow.
- **Cross-functional Collaboration:**
The data science team works with risk officers, legal advisors, and DevOps teams to ensure technical feasibility, regulatory compliance (e.g., explainability under the Equal Credit Opportunity Act), and smooth deployment.
- **Post-deployment Governance:**
Set up model monitoring dashboards, bias tracking, and retraining triggers every quarter to keep models accurate and compliant.



SCALED AGILE FRAMEWORK (SAFe)



Scaled Agile Framework (SAFe)

- The SAFe (Scaled Agile Framework) is a structured methodology for scaling Agile, Lean, and DevOps practices across large enterprises.
- SAFe is a structured framework designed to apply Agile **at scale** across multiple teams, departments, and even the entire enterprise.

<https://framework.scaledagile.com/#essential>

Alignment: Ensures all teams are working toward common strategic goals.

Faster Time-to-Market: Accelerates delivery of value through synchronized planning and execution.

Increased Productivity: Streamlines processes and improves team efficiency.

Improved Quality: Emphasizes built-in quality through continuous testing and integration.

Transparency: Provides visibility into work progress, dependencies, and risks.

Continuous Improvement: Encourages regular reflection and process enhancement.

Enhanced Collaboration: Promotes cross-functional teamwork across the organization.

Better Risk Management: Identifies and mitigates risks early through cadence and synchronization.

Customer-Centric Delivery: Focuses on delivering value that meets customer needs.

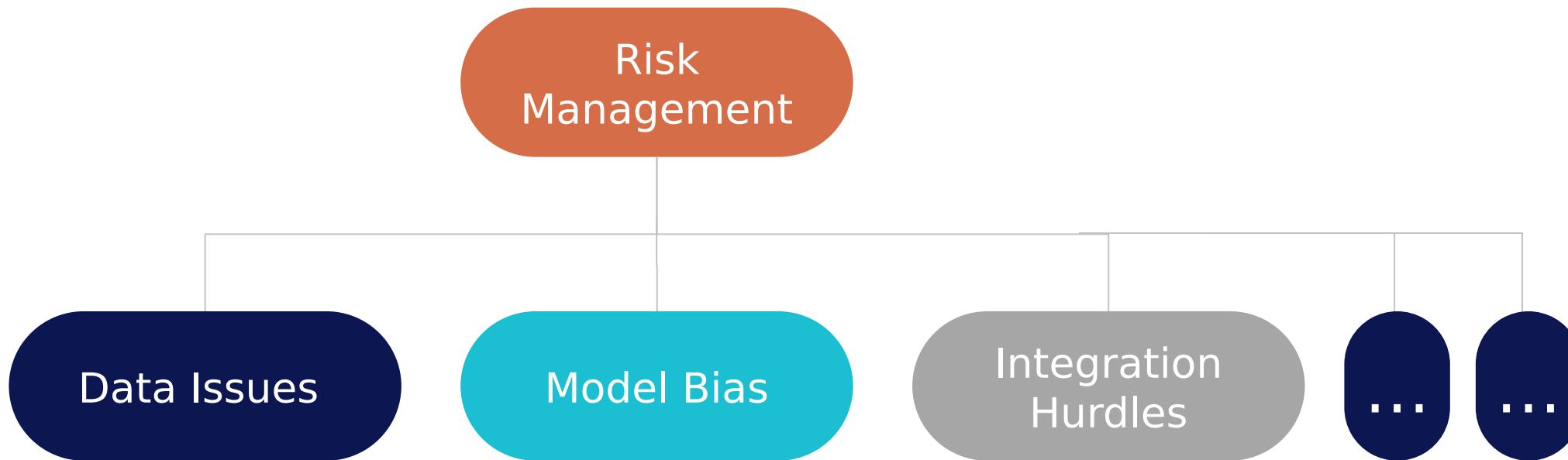
Scalability: Supports Agile implementation across large, complex enterprise environments.

RISK MANAGEMENT IN AI PROJECTS



Risk Management in AI Projects: Navigating the Unknowns

- AI projects come with unique categories of risks (e.g.: data quality, model bias, ethical concerns, and regulatory compliance) that require thoughtful planning and continuous oversight.
- Effective AI project management should balance the open-ended, iterative nature of data science with the structured, goal-oriented discipline of traditional project execution.



Common AI Project Risks & Mitigations



Category	Issue	Mitigation
Data Risks	Poor quality, insufficient quantity, inherent biases, privacy concerns.	Thorough data assessment early, data governance, ethical data sourcing, data augmentation strategies, anonymization.
Technical Feasibility Risks	Model doesn't achieve required accuracy, high computational costs, complex integration.	Start with a Proof-of-Concept, set realistic expectations, explore multiple algorithms, plan for scalability.
Stakeholder Alignment Risks	Misunderstanding of business problem, changing expectations, lack of buy-in.	Continuous communication, involve stakeholders in evaluation, clearly define success metrics, ensure executive sponsorship.
Operational & Deployment Risks	Difficult implementation, latency, scalability, model drift post-deployment, security vulnerabilities.	Implement IT/DevOps/MLOps early, design for production, plan for monitoring and retraining.
Ethical & Compliance Risks	Lack of fairness, bias, lack of transparency, regulatory non-compliance.	Establish ethical guidelines, conduct bias audits, ensure explainability where needed, consult legal/compliance teams.



KEY INGREDIENTS FOR AI PROJECT SUCCESS





AI success = technology × people × alignment



Key Success Ingredients in AI Projects



Clear Business Vision & Value Alignment: The "Why." AI must solve a real business problem and deliver measurable value.

Data Quality, Availability & Governance: Data is the lifeblood. A robust data strategy is non-negotiable.

Realistic Expectations & Iterative Progress: AI is not magic. Start small, iterate, and manage expectations.

Skilled Multidisciplinary Team & Strong Leadership: The "Who." Diverse expertise is crucial.

Effective Stakeholder Management & Communication: Keeping everyone informed and aligned.

Robust MLOps & Deployment Strategy: Moving from prototype to production reliably and scalably.

Ethical Considerations & Responsible AI: Building trust and mitigating societal risks



AI PROJECT BUDGETING



AI Projects Budgeting

- AI projects are complex and require careful financial planning. Many initiatives fail not due to poor algorithms but due to mismanaged budgets, unanticipated costs, or poor alignment with business goals.

Scoping & Planning:
Stakeholder
workshops, consultant
fees, feasibility
studies

Data Strategy:
Data acquisition,
cleaning, labeling,
storage

Model Development:
Cloud compute (GPUs,
TPUs), ML engineers,
ML tools

Experimentation &
Validation:
MLOps tools, test
datasets, model
evaluation

Deployment &
Integration:
API hosting, backend
integration, cloud
DevOps

Monitoring &
Maintenance:
Logging, retraining
cycles, system
uptime, support



Key Cost Categories



Infrastructure:
Cloud (AWS, Azure,
GCP), storage,
bandwidth.

Human Resources:
Engineers, data
scientists, PMs,
analysts.

Software/Tools:
SaaS platforms,
annotation tools,
IDEs.

Data: Licenses,
external datasets,
labeling services.

Ongoing
Operations:
Maintenance,
security, updates.



Budget Example



Category	Subcategories / Examples	Adjusted Cost Range
1. Infrastructure	GPUs, servers, local/cloud storage	\$4,000 – \$30,000 (initial setup)
2. Software & Tools	Cloud platforms, AI frameworks, dev tools	\$500 – \$7,000 + \$150-\$1,200/month
3. Talent / Human Capital	Data scientists, ML engineers, AI specialists	\$100K – \$140K/year per role (Location Dependent)
4. Training & Development	Online courses, workshops, team upskilling	\$300 – \$3,000
5. Data Acquisition	Licensed datasets, data vendors, collection software	\$1,500 – \$45,000 + \$600-\$4,500/month
6. Data Management	Cleaning, labeling, storage, security systems	\$1,200 – \$18,000 + \$150-\$2,200/month
7. Operations	Continuous data intake, model retraining, testing	\$600 – \$4,800/month
8. Maintenance	Server upkeep, cloud usage fees, cybersecurity tools	\$750 – \$9,500/month
9. Integration Costs	Custom APIs, system integration, user training	\$6,000 – \$45,000
10. Hidden/Unexpected	Advanced feature engineering, model revisions, legal/ethics review	\$3,000 – \$28,000
11. Total Project Range	Pilot to enterprise-level deployment	\$15,000 – \$450,000+



PRACTICE OPPORTUNITY



Practice Opportunity



- You are part of a consulting team advising a mid-size tech company interested in building a generative AI system. Your client wants to understand what it would take to train a model like GPT-3 or GPT-4 from scratch (or fine-tune one at scale) and whether it's financially realistic.
- Select a Frontier AI Model to Research (Choose One): GPT-3, GPT-4, Gemini 1.5, or Claude 3 (Anthropic)

Aspect	Your Research Findings
Estimated total training cost	What are the publicly available figures or best estimates?
Infrastructure & compute cost	What kind of chips (e.g., A100, V100, H100)? How many? Any cloud usage?
Talent & research personnel	What teams are typically involved (e.g., researchers, MLOps, engineers)?
Energy consumption & location	Where was it trained? What's the estimated power or energy footprint?
Ongoing operational cost	What is the estimated daily or monthly cost of inference/serving?



Practice Opportunity Solution



Category	Estimated Cost	Details & Notes
Total Training Cost	\$63M – \$100M+	Includes compute, staff, infrastructure, and energy. Sam Altman confirmed >\$100M.
GPU Hardware (CapEx)	~\$800M (purchase)	Based on NVIDIA A100 clusters (tens of thousands of GPUs); often amortized.
Amortized Hardware Cost	~\$40M	Spread out over multiple projects or usage cycles.
Compute Cost (Training)	~\$63M – \$78M	For model training using cloud or in-house clusters.
Staff & Research	~30–40% of total training cost	Includes ML engineers, researchers, alignment teams, MLOps, etc.
Interconnect & Servers	~15–22% of total training cost	Includes high-speed networking (e.g., NVLink) and non-GPU server infrastructure.
Cluster Interconnect	~9–13%	Covers networking between compute nodes.
Energy Consumption	~2–6%	Power for running clusters; varies by region and efficiency.
Water Use (Cooling)	~0.5 liters/query	For data center cooling; significant for inference at scale.
Inference (Daily Ops)	~\$700K/day	Ongoing cost to serve ChatGPT to users globally.
Cost Per Query	~\$0.0036/token (≈0.36¢/query)	Based on OpenAI's serving scale and infrastructure.

<https://www.forbes.com/sites/katharinabuchholz/2024/08/23/the-extreme-cost-of-training-ai-models/>



Summary



Start with a clear vision, business value, and feasibility check; define a focused MVP to manage scope.

Use SMART goals across model (e.g., accuracy), system (e.g., latency), and business (e.g., retention) levels.

Assign key roles: product managers, data scientists, ML engineers, MLOps, and domain experts.

Prefer Agile or Hybrid methods for flexibility; use SAFe to scale across teams.

Budget carefully and plan for risks like data quality, bias, and compliance.



Thank you!